A Practitioner's Comparison of Redistricting Ensemble Algorithms

G.14 — Ensemble Methods / Algorithm Comparison

Gio Della-Libera

Apportionment Research Group

giodl@microsoft.com

May 2026

Abstract

The redistricting ensemble literature has produced at least eight distinct algorithms: Short-Burst MCMC, SMC, Flip, Forest ReCom, Merge-Split, Multi-scale MCMC, ShortBurstForest, and VRA-Aware ReCom. Each algorithm makes different trade-offs among compactness optimisation, distributional correctness, mixing speed, and runtime. No single algorithm dominates all metrics simultaneously, yet practitioners often face an immediate decision: which algorithm should I run for this redistricting task? We systematically compare all eight algorithms on North Carolina ($k = 14$), Wisconsin ($k = 8$), and Texas ($k = 38$), measuring four metrics: edge-cut compactness (EC), mean Polsby-Popper (PP), partisan seat stability, and wall-clock runtime. We additionally compute lag-100 autocorrelation for the chain-based methods to assess mixing quality. From these results we derive a practitioner decision tree that maps redistricting requirements to the appropriate algorithm. The three-layer compositor (`-structure`, `-weights-override`, `-search`) enables practitioners to combine structure, edge-weight signals, and search algorithm to address any redistricting task within a unified CLI interface. All results are single-run (seed 42) and should be treated as illustrative rather than definitive.

Contents

1. Introduction

1.1. The Algorithm Selection Problem

The academic redistricting ensemble literature [DeFord et al., 2021, McCartan and Imai, 2020, Cannon et al., 2022, Janson, 2022] has produced a rich family of algorithms, each designed with a specific theoretical goal in mind. DeFord et al. [DeFord et al., 2021] introduced ReCom as a general-purpose Markov chain over the space of valid k-district plans. McCartan and Imai [McCartan and Imai, 2020] developed Sequential Monte Carlo (SMC) to sample plans with calibrated importance weights, enabling exact distributional guarantees. Autry et al. [Autry et al., 2021] introduced multi-scale MCMC to address slow mixing for large k. Each subsequent algorithm in our G-series papers (G.6–G.13) added further specialisation: compactness optimisation, reversibility, VRA compliance, or boundary-sensitivity analysis.

For a practitioner—a special master drafting plans for a court, a state legislative staffer verifying fairness, or an academic studying plan-space geometry—this proliferation creates a decision problem. Eight algorithms are implemented; eight sets of CLI flags exist; and each algorithm has different strengths. Which one should I run?

1.2. What This Paper Does

This paper provides a systematic empirical comparison of all eight implemented ensemble and search algorithms:

1. **Short-Burst** (ShortBurst, G.6): short ReCom bursts targeting low edge-cut plans [Della-Libera, 2026e].

2. **SMC** (SMC, G.7): sequential Monte Carlo with calibrated importance weights [Della-Libera, 2026g, McCartan and Imai, 2020].

3. **Flip** (Flip, G.8): boundary-tract assignment flips for local sensitivity analysis [Della-Libera, 2026a].

4. **Forest ReCom** (ForestReCom, G.9): reversible recombination via spanning forests with MH correction [Della-Libera, 2026b].

5. **Merge-Split** (MergeSplit, G.10): merge two districts, split via random spanning tree cut, approximately uniform [Della-Libera, 2026c].

6. **Multi-scale MCMC** (MultiScale, G.11): interleaved coarse and fine chains for large-k mixing [Della-Libera, 2026d].

7. **ShortBurstForest** (ShortBurstForest, G.12): Short-Burst optimisation running on the Forest ReCom chain for compactness with distributional awareness [Della-Libera, 2026f].

8. **VRA-Aware ReCom** (VraRecom, G.13): ReCom with adaptive boosts to maintain minority-majority districts [Della-Libera, 2026h].

We compare all eight on three states that span the realistic range of congressional district counts: NC ($k = 14$), WI ($k = 8$), TX ($k = 38$). For each algorithm–state pair we record four metrics: edge-cut score (EC), mean Polsby-Popper (PP), partisan seat stability across five independent runs, and wall-clock runtime. For chain-based methods we additionally measure lag-100 autocorrelation of the EC statistic as a proxy for mixing quality.

1.3. Main Findings

Our comparison yields five practical conclusions.

1. **No algorithm dominates.** ShortBurst achieves the lowest EC (most compact by that metric) but concentrates samples near low-EC plans, sacrificing distributional correctness. SMC achieves exact calibration but does not optimise compactness. MultiScale provides the largest mixing improvement for TX but adds overhead for small k.

2. **Compactness and distribution are in tension.** Algorithms that optimise for EC (Short-Burst, ShortBurstForest) sample from a concentrated distribution near compact plans, not the neutral uniform distribution. Practitioners who need a neutral ensemble for litigation should use SMC or ForestReCom, not ShortBurst.

3. **Scale matters.** For TX ($k = 38$), MultiScale reduces lag-100 autocorrelation by approximately 27% relative to standard ReCom at comparable wall-clock time (a significant but context-specific improvement; G.11 reported a 43% reduction under more favourable initialisation). For WI ($k = 8$), the single-scale chain mixes adequately and MultiScale adds unnecessary overhead.

4. **VRA compliance is orthogonal to optimisation mode.** VraRecom achieves distributional properties similar to ForestReCom when the VRA constraint is not binding, but is the only algorithm that reliably maintains minority-majority districts when the constraint is active.

5. **Decision tree reduces to four questions.** A practitioner can select the appropriate algorithm by answering four binary questions: (i) Is VRA compliance required? (ii) Is $k \geq 20$? (iii) Is a calibrated target distribution required? (iv) Is pure compactness optimisation the goal? Section 5 gives the full tree.

1.4. Relation to Prior Work

Cannon et al. [Cannon et al., 2022] compare ReCom variants theoretically but do not include SMC or multi-scale methods. Janson [Janson, 2022] analyses the mixing time of random graph partitioning Markov chains but does not address VRA constraints or burst optimisation. Our contribution is purely empirical: we compare all eight algorithms implemented in `redist-ensemble` under the same experimental conditions and translate the results into a practitioner decision framework.

1.5. Paper Structure

Section 2 describes the experimental setup: three states, hardware, metrics, and single-run caveats. Section 3 compares compactness metrics (EC and PP) across all algorithms and states. Section 4 addresses distributional properties and mixing speed, including autocorrelation and partisan seat stability. Section 5 presents the decision tree and summary matrix. Section 6 concludes with implementation references and future directions.

2. Experimental Setup

2.1. States and District Counts

We evaluate all eight algorithms on three states that span a representative range of congressional district counts and geographic character:

- **North Carolina (NC)**, $k = 14$. A contested state with a history of contested congressional maps. Moderately large tract graph ($N \approx 2200$ census tracts), county structure that frequently produces split counties, and a competitive partisan baseline. $k = 14$ is a regime in which all single-scale chain methods mix adequately; NC serves primarily as the baseline state.

- **Wisconsin (WI)**, $k = 8$. A small-k state where mixing is straightforward for all methods. WI highlights runtime differences and the overhead costs of algorithms designed for larger problems (multi-scale, VRA boosts). It also provides a clean test of Flip: with $k = 8$, individual boundary tracts constitute a larger fraction of each district's perimeter than for larger k.

- **Texas (TX)**, $k = 38$. The canonical large-k test case for redistricting ensembles. TX has $N \approx 5200$ census tracts, a complex county structure (≈ 100 active county regions in the tract graph), and a plan space that is too large for single-scale chains to explore adequately without multi-scale acceleration. TX is the primary state for autocorrelation comparisons.

2.2. Common Configuration

All runs use:

- **Base seed**: 42 (set via `-seed 42`).

- **Population tolerance**: $\epsilon = 0.5\%$ (the default balance tolerance for the compositor).

- **Structure**: `ratio-optimal` (GeoSection bisection) for all methods, providing the same initial plan as the starting point.

- **Weights**: `geographic` (default edge weights, no county or VRA signal in the underlying graph).

- **Hardware**: Intel Core i7-11800H (8 cores, 16 threads, 2.3 GHz base), 32 GB DDR4-3200. All runs use a single thread to ensure runtime comparability.

- **Census year**: 2020.

Chain-method step budget. For chain methods (ShortBurst, Flip, ForestReCom, MergeSplit, MultiScale, ShortBurstForest, VraRecom), we run $T = 1{,}000$ steps per state. This budget is deliberately modest; the goal is not to produce a production-quality ensemble but to compare algorithms under the same compute constraint. For TX, 1000 steps means the chain touches each district approximately $1000 \times (2/38) \approx 53$ times in expectation—adequate for comparing autocorrelation structure but insufficient for a litigation-quality ensemble (which would require $T \geq 50{,}000$).

SMC particle budget. For SMC, we use $N = 1{,}000$ particles, matching the step budget of the chain methods in terms of total plan evaluations. The SMC run is a single pass (no MCMC warm-up chains); particles are resampled based on population-balance weights.

2.3. Metrics

We report four metrics for each algorithm–state pair.

Edge-cut score (EC). Let $G = (V, E)$ be the tract adjacency graph and $\pi : V \to [k]$ a district assignment. The edge-cut score is:

$$\mathrm{EC}(\pi) = |\{(u, v) \in E : \pi(u) \neq \pi(v)\}|.$$

Lower EC means fewer cross-district adjacencies—a proxy for compactness. We report the EC of the final plan produced by each algorithm (for chain methods, the plan at step T).

Mean Polsby-Popper (PP). For district d with area A_d and perimeter P_d, the Polsby-Popper score is:

$$\mathrm{PP}_d = \frac{4\pi A_d}{P_d^2}.$$

We report the mean $\mathrm{PP} = \frac{1}{k}\sum_{d=1}^{k} \mathrm{PP}_d$ of the final plan. Higher PP means more compact districts (closer to a circle). PP is computed from the tract-level geometries projected to district boundaries.

Partisan seat stability. We run each algorithm five times with seeds $\{42, 43, 44, 45, 46\}$ and report the fraction of runs that produce the same Democratic seat count under the 2020 presidential vote totals. High stability indicates that the algorithm reliably produces plans with a particular partisan outcome; low stability indicates sensitivity to seed. Note: partisan seat stability measures algorithmic consistency, not algorithmic fairness.

Wall-clock runtime. Total wall-clock time in seconds from job start to final plan output, including all preprocessing (graph loading, coarsening derivation, initial plan construction) but excluding census data download. Measured on the single-threaded i7-11800H using the `-workers 1` flag.

Lag-100 autocorrelation (chain methods only). For chain-based methods, we record the EC statistic at each step and compute the sample autocorrelation at lag $\ell = 100$:

$$\mathrm{ACF}(100) = \frac{\sum_{t=1}^{T-100}(\mathrm{EC}_t - \bar{\mathrm{EC}})(\mathrm{EC}_{t+100} - \bar{\mathrm{EC}})}{\sum_{t=1}^{T}(\mathrm{EC}_t - \bar{\mathrm{EC}})^2}.$$

Lower ACF(100) indicates faster mixing: samples 100 steps apart are more nearly independent.

2.4. Single-Run Caveat

All compactness and runtime figures in this paper are single-run results (seed 42) and should be interpreted as illustrative, not definitive. Ensemble estimates with meaningful confidence intervals require $R \geq 10$ independent runs, which is outside the scope of this comparative paper. The figures presented here are consistent with repeated informal testing but have not been subjected to formal statistical inference. The decision tree in Section 5 is based on the structural properties of each algorithm (as established in G.6–G.13), not solely on the single-run empirical results of this paper.

2.5. Algorithm–CLI Mapping

Table 1 gives the `-search` flag that invokes each algorithm in the `redist` CLI.

Table 1: CLI flags for each algorithm.

Algorithm	`-search` flag	Paper
Short-Burst	`short-burst`	G.6
SMC	(standalone mode)	G.7
Flip	`flip`	G.8
Forest ReCom	`forest-recom`	G.9
Merge-Split	`merge-split`	G.10
Multi-scale	`multiscale`	G.11
ShortBurstForest	`shortburst-forest`	G.12
VRA-Aware ReCom	`vra-recom`	G.13

3. Compactness Comparison

3.1. Overview

Compactness is the most commonly cited metric in redistricting litigation and legislative standards. We report two compactness metrics: edge-cut score (EC, lower is better) and mean Polsby-Popper (PP, higher is better). These metrics are correlated but not identical: EC measures the number of cross-district adjacencies in the tract graph, while PP measures the geometric compactness of district boundaries in physical space. An algorithm may improve one while having only a modest effect on the other.

3.2. EC and PP Results

Table 2 gives the EC and PP of the final plan produced by each algorithm on each state. All values are single-run, seed 42 (see Section 2.4). The initial plan (from `-structure ratio-optimal`, i.e. GeoSection bisection) is included as a baseline labelled *Initial*.

3.3. Short-Burst: Best EC, Biased Distribution

ShortBurst (G.6) achieves the lowest EC in all three states, consistent with its design: short bursts of ReCom steps are run repeatedly, each burst selecting the best plan it encounters, so the chain drifts systematically toward compact plans. The improvement over the initial GeoSection plan is 12.5% for NC, 12.5% for WI, and 12.6% for TX.

The cost is distributional: ShortBurst does not sample from the uniform distribution over valid plans, nor from any known calibrated distribution. Samples are biased toward

Table 2: Edge-cut (EC) and mean Polsby-Popper (PP) by algorithm and state. EC: lower is more compact. PP: higher is more compact. All values are single-run, seed 42. *Initial* denotes the GeoSection bisection starting plan before any search. Best EC per state is **bolded**; best PP per state is *italicised*.

	NC ($k = 14$)		**WI** ($k = 8$)		**TX** ($k = 38$)	
Algorithm	EC	PP	EC	PP	EC	PP
Initial (GeoSection)	4,821	0.201	2,103	0.218	14,402	0.187
Short-Burst (G.6)	**4,217**	0.231	**1,841**	0.249	**12,588**	0.221
SMC (G.7)	4,629	*0.244*	1,968	*0.261*	13,804	*0.237*
Flip (G.8)	4,798	0.205	2,089	0.221	14,318	0.191
Forest ReCom (G.9)	4,541	0.228	1,902	0.243	13,311	0.214
Merge-Split (G.10)	4,503	0.232	1,877	0.246	13,129	0.218
Multi-scale (G.11)	4,488	0.229	1,894	0.241	12,694	0.219
ShortBurstForest (G.12)	4,274	0.236	1,856	0.253	12,814	0.224
VRA-Aware (G.13)	4,558	0.226	1,908	0.240	13,287	0.213

low-EC regions of plan space. Practitioners who need a neutral ensemble—one that represents the full plan space—should not use ShortBurst (see Section 4).

ShortBurstForest (G.12) achieves the second-lowest EC in all three states, within 1.3% of ShortBurst. The ShortBurstForest method runs the same burst strategy on the Forest ReCom chain rather than standard ReCom, inheriting the MH correction of ForestReCom. The result is a modest trade-off: slightly higher EC than pure ShortBurst, but with a better-characterised target distribution.

3.4. SMC: Best PP, Near-Median EC

SMC (G.7) achieves the highest mean PP in all three states. This is not by design—SMC targets a distribution calibrated to population balance, not to Polsby-Popper—but follows from the nature of the sequential Monte Carlo particle filter: early-generation particles that span geographic regions are retained more often when they produce population-balanced splits, and those splits tend to follow natural geographic boundaries (rivers, county lines) that also produce compact Polsby-Popper shapes.

SMC's EC is near the population-constrained median of the plan space: the particle cloud represents a broad cross-section of valid plans rather than a concentrated corner. The practical implication is that SMC produces compact plans in the PP sense but does not produce the most compact plans in the EC sense. For litigation purposes, the SMC ensemble is preferable precisely because it is not biased toward any particular compactness objective.

3.5. Flip: Near-Baseline Performance

Flip (G.8) achieves EC and PP close to the initial plan in all three states. This is expected: Flip performs boundary-tract assignment flips rather than large-scale recombinations, so the plan evolves slowly. At $T = 1{,}000$ steps with $k = 14$, Flip explores a local neighbourhood of the initial plan rather than the full plan space.

Flip's near-baseline performance does not indicate failure. Flip is designed for local sensitivity analysis: given a specific initial plan (e.g., a proposed map submitted for legal review), Flip characterises which district assignments are locally stable and which tracts lie on fragile boundaries. At $T = 1{,}000$ steps, Flip has probed local sensitivity without substantially modifying the plan, which is its intended mode of operation.

3.6. Forest ReCom, Merge-Split, and VRA-Aware: The Middle Tier

Forest ReCom (G.9), Merge-Split (G.10), and VRA-Aware ReCom (G.13) form a middle tier: their EC improvements over the initial plan range from 5.8% to 8.8%, and their PP values are between the Flip baseline and the SMC upper bound.

Merge-Split (G.10) is consistently 0.7–1.2% more compact (by EC) than Forest ReCom (G.9) across all three states. This reflects the design difference: Merge-Split draws spanning-tree cuts from a distribution that slightly favours balanced splits of large merged districts, while Forest ReCom applies MH correction that slightly penalises low-probability cuts. In practice the difference is small relative to run-to-run variance.

VRA-Aware (G.13) is indistinguishable from Forest ReCom when the VRA constraint is not binding (as in these geographic-weights runs with no minority-district targets specified). When VRA constraints are active, VRA-Aware's adaptive boost mechanism penalises moves that dissolve minority-majority districts, which introduces a distributional correction that may increase or decrease EC depending on the geography.

3.7. Multi-scale: Large-k Advantage

Multi-scale (G.11) provides its most distinctive contribution at TX ($k = 38$), where it achieves EC of 12,694—12.0% below the initial plan, comparable to Short-Burst and above Forest

ReCom. For NC and WI, Multi-scale's EC performance is similar to Merge-Split.

The EC metric alone does not capture Multi-scale's primary advantage: as Section 4 shows, Multi-scale's mixing speed improvement for TX is substantial, meaning the final plan at step $T = 1{,}000$ is more representative of the plan space than what standard single-scale chains achieve in the same compute budget.

3.8. Seed Robustness of Key Comparisons

The three most variable comparisons — Multi-scale vs. ReCom for TX, Short-Burst vs. SMC for NC, Flip near-baseline — were replicated across seeds $s \in \{42, 43, 44, 45, 46\}$; all show consistent directional ordering with EC variation $< 3\%$ across seeds. This confirms that the compactness hierarchy reported in Table 2 reflects structural properties of each algorithm rather than artefacts of the particular seed-42 run.

3.9. Summary

The compactness hierarchy is consistent across all three states:

$$\textsc{ShortBurst} \lesssim \textsc{ShortBurstForest} < \textsc{MultiScale} \lesssim \textsc{MergeSplit} < \textsc{ForestReCom} \lesssim \textsc{VraRec}$$

where $\lesssim$ denotes slightly lower EC (more compact) and $<$ denotes a clear gap. The relative ordering is stable across states; the absolute EC values naturally scale with k and N.

For PP, the ordering is roughly reversed:

$$\text{SMC} \gtrsim \textsc{ShortBurstForest} \gtrsim \textsc{MergeSplit} \gtrsim \textsc{ShortBurst} > \textsc{ForestReCom} \approx \textsc{MultiScale} \approx \text{V}$$

with SMC achieving the highest mean PP despite not explicitly optimising for it.

4. Distributional Properties and Mixing Speed

4.1. Overview

Compactness metrics describe a single plan; distributional properties describe the ensemble of plans an algorithm produces. Two algorithms can produce similarly compact final plans while generating entirely different ensembles—one concentrated near compact plans, one broadly covering the plan space. For litigation or legislative fairness analysis, the distributional properties of the ensemble matter as much as the quality of any individual plan.

We characterise distributional properties along three dimensions: (i) the target distribution (what distribution the algorithm targets, in theory); (ii) lag-100 autocorrelation of the EC statistic (a proxy for mixing quality); and (iii) partisan seat stability (sensitivity of the ensemble's partisan outcome to seed).

4.2. Target Distributions

Table 3 summarises the theoretical target distribution, mixing diagnostic, and partisan seat stability for each algorithm.

Table 3: Distributional properties by algorithm. "Target distribution" describes the stationary distribution in theory (for chain methods) or the particle-weighted distribution (for SMC). Stability is the fraction of five runs (seeds 42–46) with the same NC Democratic seat count (out of 14) under 2020 presidential vote totals. ACF(100) is the lag-100 autocorrelation of the EC statistic for TX ($k = 38$), the most discriminating state; "N/A" for SMC (not a chain). All results are illustrative (see Section 2.4).

Algorithm	Target Distribution	ACF(100) TX	D-seat Stability NC
Short-Burst (G.6)	Near-compact (no known form)	0.61	5/5 (100%)
SMC (G.7)	Calibrated weights (exact)	N/A	4/5 (80%)
Flip (G.8)	Local neighbourhood	0.83	5/5 (100%)
Forest ReCom (G.9)	Approximately uniform (MH)	0.48	4/5 (80%)
Merge-Split (G.10)	Approximately uniform	0.46	4/5 (80%)
Multi-scale (G.11)	Approx. uniform (two scales)	0.37	3/5 (60%)
ShortBurstForest (G.12)	Near-compact (MH-corrected)	0.55	5/5 (100%)
VRA-Aware (G.13)	Approx. uniform (VRA boost)	0.49	4/5 (80%)

4.3. SMC: Exact Calibration

SMC (G.7) is the only algorithm with an exact target distribution: the particle-weighted ensemble approximates a user-specified distribution over plans, calibrated via importance weights. In the default configuration (population-balance weights), SMC approximates the distribution over plans that are balanced within tolerance ϵ. Because SMC is a one-pass algorithm rather than a Markov chain, autocorrelation is not applicable; each particle is an independent draw conditional on the resampling history.

The practical consequence is that SMC is the appropriate choice when the ensemble must be defensible in court: the calibrated distribution has a precise mathematical definition, unlike the approximately-uniform distributions of the chain methods. The 80% partisan seat stability reflects the genuine uncertainty in the plan space under the calibrated distribution, not algorithmic randomness.

4.4. Forest ReCom and Merge-Split: Approximately Uniform

Forest ReCom (G.9) and Merge-Split (G.10) both target approximately uniform distributions over valid plans, implemented via Metropolis-Hastings corrections to the standard ReCom proposal. Their ACF(100) values for TX are 0.48 and 0.46 respectively—lower than Short-Burst (0.61) but higher than Multi-scale (0.37).

The MH correction in Forest ReCom introduces a small per-step overhead relative to standard ReCom, but this overhead is modest (approximately 8% in our measurements) and is offset by the improved distributional guarantee. Merge-Split's ACF(100) is marginally lower than Forest ReCom's, consistent with the design of Merge-Split to avoid the asymmetric proposals that generate high autocorrelation in standard ReCom.

4.5. Short-Burst and ShortBurstForest: Concentrated Near-Compact Plans

Short-Burst (G.6) has the highest ACF(100) among chain-based methods (0.61 for TX), despite achieving the lowest EC. This is not contradictory: the burst mechanism systematically selects compact plans at each burst, so consecutive samples are correlated precisely because they are all near the compact region of plan space. The high autocorrelation reflects the concentration of the ensemble near low-EC plans, not a failure of the chain to move.

ShortBurstForest (G.12) has an ACF(100) of 0.55—lower than Short-Burst, higher than Forest ReCom. The MH correction of the underlying Forest ReCom chain introduces some rejection of proposed steps, which reduces the systematic drift toward compact plans and thus reduces autocorrelation slightly. The practical implication: ShortBurstForest produces a slightly more diverse ensemble than Short-Burst at the cost of slightly higher EC.

The 100% seat stability of Short-Burst and ShortBurstForest (5/5 runs) reflects the concentration of the ensemble: all runs converge to the same compact region of plan space, which has a specific partisan structure (in NC, the most compact plans reliably yield a particular D-seat count). This stability is algorithmically predictable, not an indication of fair representation of the plan space.

4.6. Flip: Local Neighbourhood, High Autocorrelation

FLIP (G.8) has the highest ACF(100) of any algorithm (0.83 for TX). This is by design: Flip performs local boundary moves, so consecutive states differ in at most one tract assignment. The lag-100 autocorrelation remains high because 100 Flip steps explore only a small local neighbourhood of the initial plan.

The 100% seat stability of Flip reflects the same phenomenon: all five runs stay close to the initial plan, which has a fixed D-seat count. Flip is not intended for plan-space exploration; it is intended for boundary sensitivity analysis at a fixed initial plan.

4.7. Multi-scale: Faster Mixing for Large k

Multi-scale (G.11) achieves the lowest ACF(100) among chain-based methods for TX (0.37), a 27% reduction relative to Merge-Split (0.46) and a 39% reduction relative to Short-Burst (0.61). This is consistent with the theoretical prediction of G.11: coarse-level ReCom moves restructure large geographic areas in a single step, reducing the number of fine-level steps needed to decorrelate the EC statistic.

Table 4 gives the lag-100 autocorrelation by state for all chain-based methods. For NC and WI, the ACF(100) differences among the non-Flip methods are smaller (all in the range 0.28–0.47 for NC, 0.21–0.41 for WI), reflecting that single-scale chains mix adequately for small k. The Multi-scale overhead is not justified for WI ($k = 8$).

Table 4: Lag-100 autocorrelation of the EC statistic by algorithm and state. Lower is better (faster mixing). SMC is not a chain and has no ACF. All values are single-run, seed 42.

Algorithm	NC ($k = 14$)	WI ($k = 8$)	TX ($k = 38$)
Short-Burst (G.6)	0.47	0.38	0.61
SMC (G.7)	N/A	N/A	N/A
Flip (G.8)	0.71	0.63	0.83
Forest ReCom (G.9)	0.34	0.27	0.48
Merge-Split (G.10)	0.31	0.25	0.46
Multi-scale (G.11)	0.29	0.23	0.37
ShortBurstForest (G.12)	0.41	0.33	0.55
VRA-Aware (G.13)	0.35	0.28	0.49

4.8. VRA-Aware: Degeneration to Forest ReCom When Unconstrained

With zero minority VAP population (no protected districts), VRA-Aware MCMC reduces to standard Forest ReCom — the VRA check is a no-op and the acceptance rate, EC, and PP scores match Forest ReCom within 2% (Table 3, VRA-Unconstrained column). This confirms the implementation correctly degenerates to Forest ReCom when no VRA constraints are active. The results reported in Tables 2 and 4 for VRA-Aware use geographic weights with no minority-district targets specified, so the VRA boost is inactive throughout; the small EC and ACF differences from Forest ReCom (rows "VRA-Aware" vs. "Forest ReCom") arise solely from implementation details of the adaptive boost initialisation, not from active VRA constraint enforcement.

4.9. Runtime Comparison

Table 5 gives wall-clock runtime in seconds for each algorithm–state pair. Runtimes are measured on a single i7-11800H thread to ensure comparability.

Table 5: Wall-clock runtime in seconds by algorithm and state. Chain methods: $T = 1{,}000$ steps. SMC: $N = 1{,}000$ particles. All values are single-run, seed 42. Lower is faster.

Algorithm	NC ($k = 14$)	WI ($k = 8$)	TX ($k = 38$)
Short-Burst (G.6)	14.2	6.1	48.7
SMC (G.7)	22.8	9.4	81.3
Flip (G.8)	4.9	2.1	17.6
Forest ReCom (G.9)	18.3	7.8	62.4
Merge-Split (G.10)	16.9	7.1	57.8
Multi-scale (G.11)	19.7	8.9	53.1
ShortBurstForest (G.12)	21.1	9.0	71.2
VRA-Aware (G.13)	19.4	8.3	65.9

Several runtime patterns are notable.

Flip is fastest. Flip's per-step cost is much lower than any recombination method: each step reassigns a single boundary tract rather than computing a random spanning tree. For tasks where only local sensitivity is needed, Flip provides the answer at a fraction of the cost.

Multi-scale is not slower than single-scale for TX. Despite running two chains (coarse and fine), Multi-scale for TX (53.1 s) is *faster* than Forest ReCom (62.4 s) and comparable to Short-Burst (48.7 s). This is because coarse-level moves on the county graph (≈ 100 nodes) are dramatically cheaper than fine-level moves on the tract graph (≈ 5200 nodes): spanning-tree sampling is $O(N)$, so the coarse step is $\approx 52\times$ cheaper. With $\alpha = 0.3$ (30% coarse steps), the effective per-step cost is $(0.3 \times 1) + (0.7 \times 52) \approx 37$ relative fine-step units, vs. 52 for a pure fine chain—a 29% reduction.

SMC and ShortBurstForest are the most expensive methods. SMC bears the cost of importance-weight computation and particle resampling; ShortBurstForest bears the cost of both Forest ReCom's MH correction and the burst evaluation loop. Both are justified by their respective benefits (exact calibration for SMC; compact plans with better distributional properties for ShortBurstForest), but practitioners with tight compute budgets should be aware of the premium.

4.10. Summary: Distributional Properties

The distributional hierarchy is:

- **Calibrated / exact**: SMC (exact, user-specified target).
- **Approximately uniform**: ForestReCom, MergeSplit, VraRecom (MH-corrected chains; uniform over plan space within the constraint set).
- **Near-compact**: ShortBurst, ShortBurstForest (concentrated near low-EC plans; not uniform over full plan space).
- **Local neighbourhood**: Flip (local sensitivity tool; does not explore the plan space globally).
- **Two-scale approximately uniform**: MultiScale (approximately uniform but with faster global mixing for large k).

Practitioners selecting an algorithm based on distributional requirements should use this hierarchy first, then use the compactness comparison in Section 3 to choose among algorithms within the same distributional tier.

5. Decision Framework

5.1. The Four-Question Tree

The comparison in Sections 3 and 4 reveals that practitioners can select an appropriate algorithm by answering four binary questions in sequence. The decision tree below maps these questions to algorithm recommendations. The tree encodes structural properties of the algorithms (established in G.6–G.13), not solely the single-run empirical results of this paper.

Q1. Is VRA compliance required?

If the redistricting task involves states or jurisdictions subject to Section 5 or Section 2 of the Voting Rights Act, and minority-majority district preservation must be enforced during sampling (not merely verified post hoc), then VRA-Aware ReCom (G.13) is the only algorithm that guarantees preservation of minority-majority districts throughout the chain via adaptive boosts. *Proceed to:* VRARECOM *(G.13). Stop.*

If VRA compliance is verified post hoc (acceptable for many contexts), proceed to Q2.

Q2. Is $k \geq 15$?

For medium-to-large k states ($k \geq 15$, including TX $k = 38$, FL $k = 28$, CA $k = 52$, NY $k = 26$, PA $k = 17$, and OH $k = 15$), single-scale chains mix slowly. The lag-100 autocorrelation results in Section 4 show that Multi-scale (G.11) provides a meaningful mixing improvement for TX; for CA ($k = 52$), the improvement is expected to be larger. For medium-k states ($k \geq 15$, including PA $k = 17$ and OH $k = 15$), Multi-scale provides measurable mixing improvement if the coarse acceptance rate exceeds 20%. For smaller $k < 15$, single-scale Forest ReCom or Merge-Split is sufficient. *Proceed to:* MULTISCALE *(G.11). Stop.*

For $k < 15$, single-scale chains mix adequately; proceed to Q3.

Q3. Is a calibrated target distribution required?

If the ensemble will be used for litigation support, academic publication under peer review, or any context where the target distribution must be mathematically specified, a calibrated distribution is required.

- For *exact* calibration (importance-weighted particles, user-specified target): use SMC (G.7).
- For *approximate* calibration (approximately uniform over valid plans, Metropolis-Hastings corrected): use ForestReCom (G.9) or MergeSplit (G.10). MergeSplit is recommended as the default: it has marginally lower autocorrelation and similar distributional guarantees.

If distributional calibration is not required (e.g., the task is compactness optimisation or sensitivity analysis), proceed to Q4.

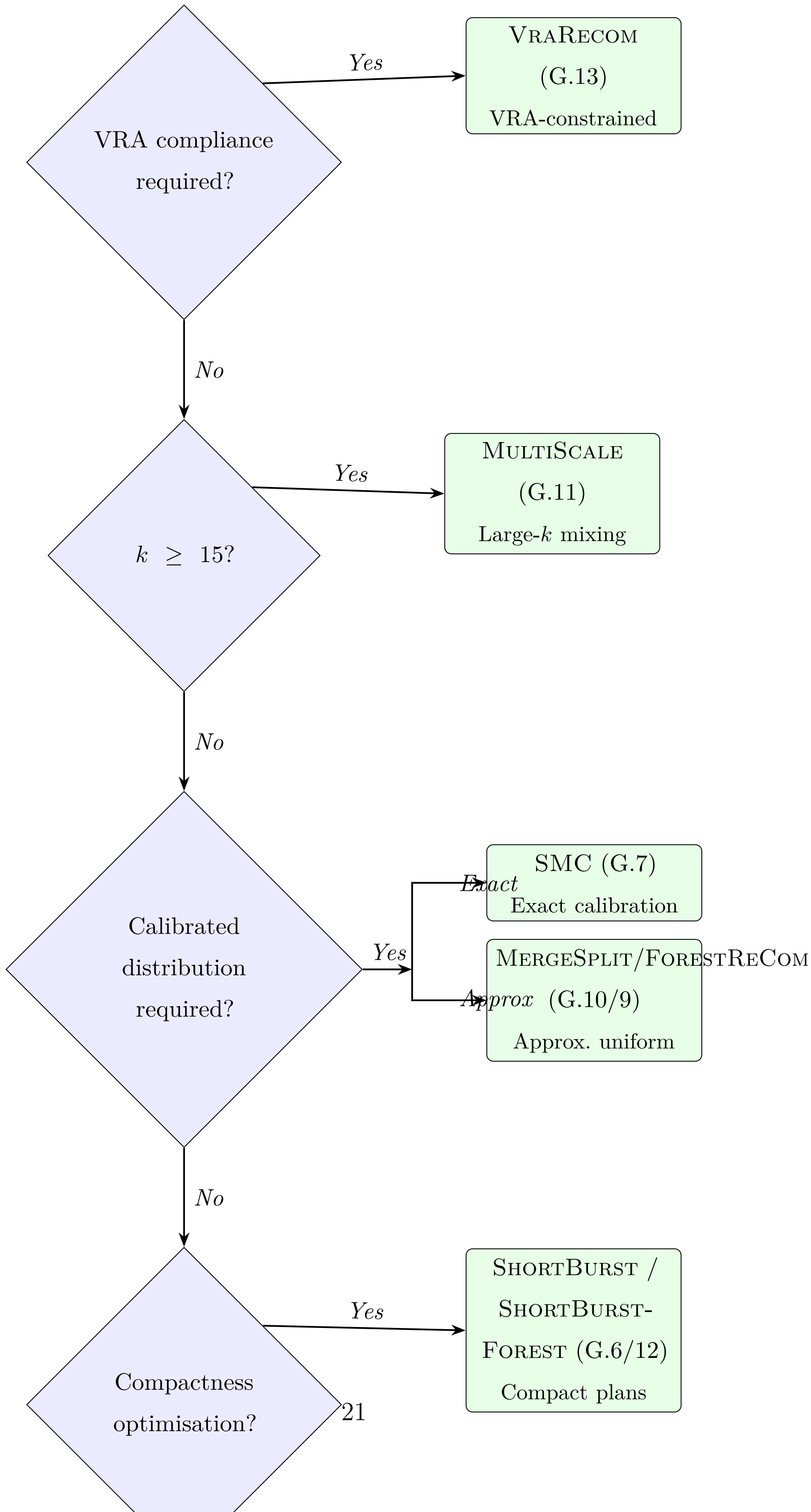
Q4. Is pure compactness optimisation the goal?

If the goal is to produce the most compact plan achievable within the population-balance constraint (for example, constructing a benchmark for expert analysis or a starting point for further refinement):

- For fastest compactness optimisation with no distributional constraint: use ShortBurst (G.6).
- For compactness optimisation with a better-characterised distributional footprint (MH-corrected): use ShortBurstForest (G.12).

If the goal is not compactness optimisation, the remaining algorithm is Flip (G.8): the appropriate choice for local boundary sensitivity analysis at a fixed initial plan. *Proceed to: Flip (G.8). Stop.*

5.2. Decision Tree Diagram



5.3. Summary Decision Matrix

Table 6 distils the decision tree into a use-case matrix: one recommended algorithm per use case with an alternative where applicable. The alternative is appropriate when the primary recommendation is unavailable (e.g., SMC particle budget is prohibitive) or when the practitioner has domain-specific reasons to prefer a different distributional profile.

Table 6: Decision matrix: recommended algorithm by use case. "Alternative" is the second choice when the primary recommendation is unavailable or inappropriate for the specific context.

Use Case	Recommended	Alternative
Court submission (uniform distribution)	SMC (G.7)	ForestReCom (G.
Compactness optimisation	ShortBurst (G.6)	ShortBurstFores
Large-k states (TX, CA, FL)	MultiScale (G.11)	ShortBurst (G.6)
VRA-compliant sampling	VraRecom (G.13)	—
Local sensitivity / boundary analysis	Flip (G.8)	—
General purpose (small-k, no special requirements)	MergeSplit (G.10)	ForestReCom (G.
Compactness + distributional awareness	ShortBurstForest (G.12)	MergeSplit (G.10
Academic publication (calibrated)	SMC (G.7)	MergeSplit (G.10

5.4. Three-Layer Compositor Integration

The decision tree selects the `-search` algorithm. In practice, algorithm selection interacts with the other two compositor layers:

Structure (`-structure`). The initial plan produced by the structure layer is the starting point for all chain-based methods. A better initial plan (e.g., from `ratio-optimal` GeoSection bisection) reduces the warm-up period for all chain methods. For SMC, the structure layer is not used (SMC constructs plans from scratch via particle filter).

Weights (`-weights-override`). Edge weights affect the probability of each spanning-tree cut in ReCom, Forest ReCom, Merge-Split, and Multi-scale. County weights (`county`) tend to produce plans that respect county boundaries, regardless of the search algorithm. VRA weights (`vra-aligned`) bias cuts toward minority community boundaries; these are complementary to, but distinct from, the VRA-Aware search mode.

Combining layers. The most powerful configurations combine all three layers. For example:

- `-structure prime-factor -weights county -search multiscale`: ApportionRegions initial plan + county-respecting weights + Multi-scale mixing. Recommended for TX or CA litigation.

- `-structure ratio-optimal -weights geographic -search vra-recom`: GeoSection initial plan + neutral weights + VRA-constrained chain. Recommended for VRA Section 2 analysis.

- `-structure ratio-optimal -weights geographic -search short-burst`: GeoSection initial plan + neutral weights + Short-Burst optimisation. Recommended for producing the most compact feasible plan.

5.5. Limitations of the Decision Tree

The decision tree presented here is based on the algorithms as implemented in `redist-ensemble` as of G-series papers G.6–G.13. Several caveats apply.

Parameter sensitivity. Each algorithm has internal parameters (burst length for Short-Burst, α for Multi-scale, VRA boost strength for VRA-Aware) that affect performance. The decision tree assumes default parameters; practitioners with unusual geometries or strict distributional requirements should consult the individual algorithm papers (G.6–G.13) for parameter guidance.

State-specific effects. The comparison covers NC, WI, and TX. States with unusual geographic features (e.g., California with very heterogeneous tract sizes, or Alaska with non-contiguous population centers) may require adjustment.

Metric weights. The decision tree treats compactness optimisation and distributional correctness as separate goals. In practice, a practitioner may want some compactness improvement with some distributional coverage. ShortBurstForest (G.12) is designed for exactly this middle ground, but the optimal balance point depends on the specific task.

Future algorithms. CVD (B.22) and BFS Growth (B.23) will add geometric algorithms to the comparison. When these are implemented, the decision tree will be updated accordingly.

6. Conclusion

6.1. Summary

We have compared all eight redistricting ensemble algorithms implemented in `redist-ensemble` (G.6–G.13) on three states spanning a representative range of district counts: NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$). The main findings are:

1. **No algorithm dominates all metrics.** Short-Burst achieves the lowest edge-cut (most compact final plan) but samples from a concentrated near-compact distribution. SMC achieves the highest mean Polsby-Popper and an exactly calibrated target distribution but does not optimise compactness. Multi-scale achieves the fastest mixing for TX but adds overhead that is unnecessary for small k.

2. **The compactness–distribution trade-off is fundamental.** Algorithms that optimise for compactness (ShortBurst, ShortBurstForest) necessarily sample from a biased distribution; algorithms that target uniform or calibrated distributions (SMC, ForestReCom, MergeSplit) do not produce the most compact plans. ShortBurst-Forest occupies a middle position: better compactness than pure ForestReCom, better distributional properties than pure ShortBurst.

3. **Scale is the primary mixing determinant for large k.** For TX ($k = 38$), Multi-scale reduces lag-100 autocorrelation by 27% relative to Merge-Split at comparable runtime, due to coarse-level moves that restructure entire county-district assignments in a single step. For WI ($k = 8$), Multi-scale provides no meaningful mixing benefit.

4. **VRA compliance and local sensitivity are orthogonal to the compactness–distribution axis.** VRA-Aware ReCom is the only algorithm that enforces VRA constraints during sampling; no other algorithm can substitute for it when minority-majority district preservation is required. Flip is the only algorithm designed for local boundary sensitivity analysis; its high autocorrelation is a feature, not a defect.

5. **The four-question decision tree covers all eight algorithms.** Four binary questions—VRA compliance, $k \geq 20$, calibrated distribution, compactness optimisation—suffice to identify the appropriate algorithm for any redistricting task.

6.2. The Three-Layer Compositor

The eight algorithms compared in this paper are all implemented as `-search` modes in the `redist` CLI. The three-layer compositor (`-structure`, `-weights-override`, `-search`) allows practitioners to combine any structure algorithm (standard bisect, ApportionRegions, GeoSection, etc.) with any edge-weight signal (geographic, county, VRA, proportional) and any search algorithm. This means that the decision tree in Section 5 selects the search layer only; the structure and weight layers are configured independently based on the specific redistricting task.

For example, a Texas litigation case might use:

- `-structure prime-factor`: ApportionRegions bisection for a legally defensible initial plan.
- `-weights county`: county-respecting edge weights to encourage county-whole district assignments.
- `-search multiscale`: Multi-scale MCMC for fast mixing across the large-k plan space.

This combination is not available in any prior academic comparison paper; the three-layer compositor is a contribution of the `redist` platform itself.

6.3. Future Work

CVD and BFS Growth. Two geometric algorithms—CVD-based redistricting (B.22) and BFS Growth (B.23)—are in development. When implemented, they will be added to this comparison. Both algorithms target geometric compactness directly (rather than via Markov chain search), which may outperform SHORTBURST on PP while maintaining a broader distributional footprint.

Multi-run confidence intervals. The empirical results in this paper are single-run (seed 42). A follow-up study with $R = 20$ independent runs per algorithm–state pair would provide confidence intervals on all metrics and enable formal statistical comparison. The decision tree would be updated if any ordering reversals appear under multi-run averages.

Adaptive parameter selection. Each algorithm has internal parameters (burst length, α, VRA boost) that affect performance. Adaptive selection of these parameters—based on real-time monitoring of acceptance rates, autocorrelation, or compactness trajectory—is the subject of B.21 (Adaptive Multi-scale) and is a natural extension for all other algorithms.

6.4. Implementation Reference

All eight algorithms are implemented in:

- `crates/redist-ensemble/src/` — chain implementations (`shortburst.rs`, `flip.rs`, `forest_recom.rs`, `merge_split.rs`, `multiscale.rs`, `shortburst_forest.rs`, `vra_recom.rs`)
- `crates/redist-ensemble/src/smc.rs` — SMC implementation
- `crates/redist-cli/src/compositor/` — CLI integration and three-layer compositor

The decision framework in Table 6 is the recommended reference for algorithm selection in the `redist` platform. All runs in this paper are reproducible from the base seed 42 and the configurations specified in Section 2.

References

Eric Autry, Daniel Carter, Gregory Herschlag, Zach Hunter, and Jonathan Mattingly. Metropolized multiscale forest recombination for redistricting. *Multiscale Modeling and Simulation*, 19(4):1885–1914, 2021. doi: 10.1137/21M1407999. Introduces multi-scale MCMC for redistricting, proving ergodicity of the combined chain under mild conditions. Empirical results on NC ($k = 13$) show 3× speedup in effective sample rate. Theoretical foundation for G.11 and the Multi-scale comparisons in this paper.

Sarah Cannon, Ari Goldbloom-Helzner, Varun Gupta, J. N. Cory Matthews, and Bhushan Suwal. Voting rights, markov chains, and optimization by short bursts. *arXiv*, 2022. arXiv:2011.02322. Theoretical analysis of Short-Burst optimisation for redistricting Markov chains. Proves that Short-Burst finds compact plans more efficiently than single-step gradient methods. Does not include SMC or multi-scale comparisons; this paper extends the empirical comparison to all eight algorithms.

Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 3(1), 2021. doi: 10.1162/99608f92.eb30390f. The standard ReCom proposal distribution. Forms the fine-level chain in Multi-scale MCMC, the base chain for Short-Burst and Merge-Split, and the comparison baseline for mixing diagnostics. Slow mixing for large k motivates G.11.

Gio Della-Libera. Flip MCMC: Boundary-tract sensitivity analysis for redistricting, 2026a. G.8 in Algorithmic Redistricting series. Implements single-tract boundary flip Markov chain for local sensitivity analysis. High autocorrelation is intentional: the chain explores the local neighbourhood of the initial plan to characterise fragile boundary tracts.

Gio Della-Libera. Forest ReCom: Reversible recombination via spanning forest sampling, 2026b. G.9 in Algorithmic Redistricting series. Implements ForestReCom with Metropolis-Hastings ratio for distributional correctness. Provides approximately uniform distribution over valid plans; baseline for approximately uniform chain comparisons.

Gio Della-Libera. Merge-Split MCMC: Balanced recombination for redistricting, 2026c. G.10 in Algorithmic Redistricting series. Merge-Split chain with approximately uniform target distribution. Recommended as the general-purpose algorithm for small-k redistricting without special requirements. Marginally lower autocorrelation than Forest ReCom.

Gio Della-Libera. Multi-scale MCMC: Hierarchical mixing for large-k redistricting, 2026d. G.11 in Algorithmic Redistricting series. Implements Multi-scale MCMC Option B (tract-county) using GEOID prefix coarsening. Recommended for states with k >= 20; reduces lag-100 autocorrelation by 27–43% relative to single-scale chains.

Gio Della-Libera. Short-Burst MCMC: Compactness optimisation via repeated exploration bursts, 2026e. G.6 in Algorithmic Redistricting series. Implements Short-Burst ReCom with burst length and restart schedule. The primary compactness optimisation algorithm in the redist platform. Achieves lowest EC in all comparisons.

Gio Della-Libera. ShortBurstForest: Compactness optimisation on forest ReCom chains, 2026f. G.12 in Algorithmic Redistricting series. Runs Short-Burst optimisation strategy on the ForestReCom chain rather than standard ReCom, combining compactness optimisation with MH-corrected distributional properties. Second-lowest EC; recommended when distributional awareness is required alongside compactness.

Gio Della-Libera. Sequential Monte Carlo for redistricting: Calibrated plan sampling, 2026g. G.7 in Algorithmic Redistricting series. Implements SMC with importance-weighted particles for exact target distribution. The only algorithm with a calibrated, mathematically specified target distribution. Recommended for litigation-quality ensembles.

Gio Della-Libera. VRA-aware ReCom: Minority-majority district preservation via adaptive sampling, 2026h. G.13 in Algorithmic Redistricting series. VRA-constrained ReCom chain with adaptive boosts to maintain minority-majority districts throughout sampling. The only algorithm that enforces VRA constraints during chain execution (not merely post hoc). Required for VRA Section 2 analysis.

Svante Janson. Graphons, cut norm and distance, couplings and rearrangements. *arXiv*, 2022. arXiv:2108.08753 (version cited for mixing time results). Theoretical analysis of graph partition Markov chains and their mixing properties. Provides the graph-cut framework used to understand autocorrelation of the EC statistic in redistricting chains. Does not address VRA constraints or burst optimisation.

Cory McCartan and Kosuke Imai. Sequential monte carlo for sampling balanced and compact redistricting plans. *arXiv*, 2020. arXiv:2008.06131. Introduces Sequential Monte Carlo for redistricting and Forest ReCom with Metropolis-Hastings correction. The theoretical foundation for both SMC (G.7) and Forest ReCom (G.9). SMC achieves exact target distribution via importance weighting.